Week 1

Why This Course Exists

Session 00

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How the course runs

5 weeks. 10 sessions.

Two 1.5-hour sessions every Thursday, back to back

Between sessions

Weekly lecture notes (and more) on GitHub.

No grades. Optional homework.

Prerequisites at a working level

Linear algebra	Vectors, matrices, change of basis, eigenvalues and eigenvectors.
Calculus	Derivatives, gradients, chain rule and basic optimization.
Probability & statistics	Expectation, variance, conditional probability and estimation.
Python	NumPy, Matplotlib and Jupyter notebooks.

The purpose of this course

A working pipeline can hide a poorly formulated problem.

The central question is whether the model’s assumptions match the structure of the phenomenon.

PART 1

A conventional ML pipeline

What would you try when the measurements are all you have?

PART 2

What is machine learning?

A big picture scenario.

What is machine learning?

Machine learning is the process of learning useful ***structure*** from data. We observe some inputs and outputs, and we want a program that maps new inputs to correct outputs.

What is machine learning?



What is machine learning?





What is machine learning?

Analogy. Classical programming is like giving someone a recipe. Machine learning is like giving them a *kitchen*, *ingredients*, and *a way to measure* whether the *food* tastes good—and letting them figure out the recipe themselves.

The components of learning

Hypothesis space F	The functions the model is allowed to express.
Dataset D	The observed input–output pairs (x_i, y_i).
Loss L	A measure of prediction error.
Optimization	The search for parameters that reduce the loss.

Newton was doing machine learning!



$$F = ma$$

PART 3

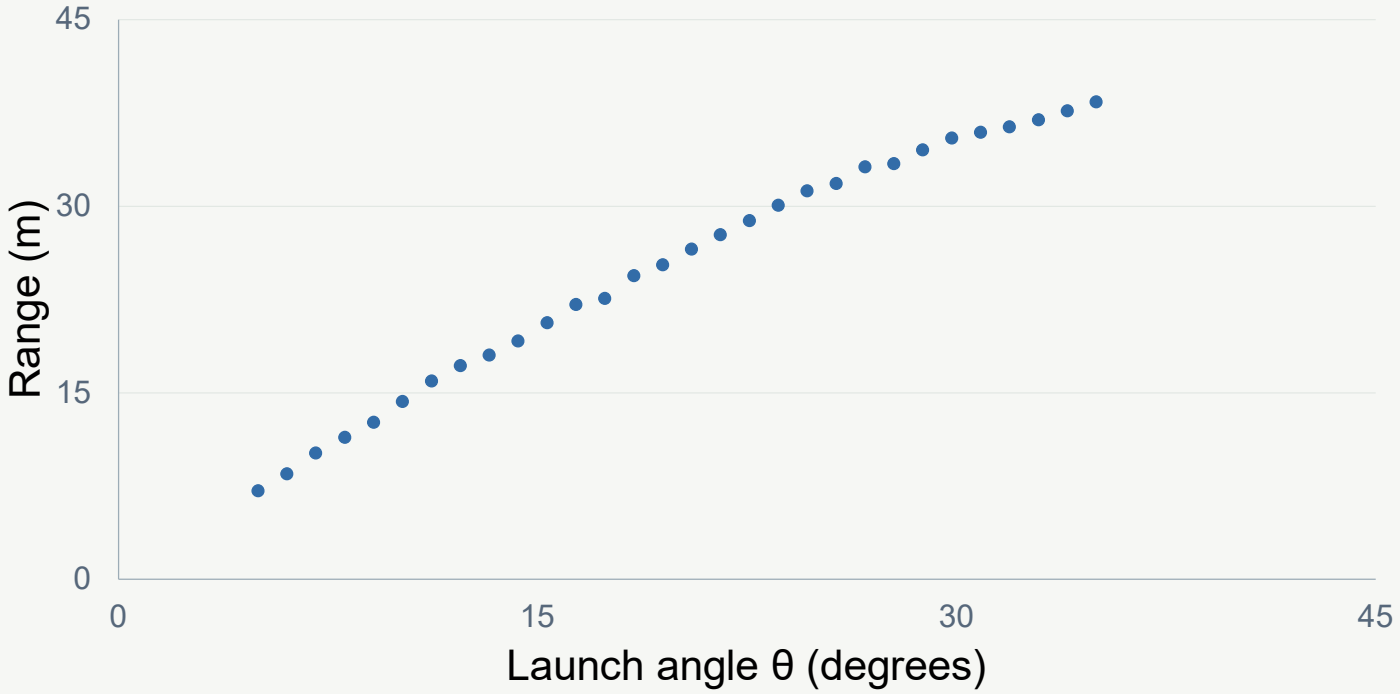
A conventional ML pipeline, results.

The prediction task

Physical setup	Launch speed v. Level ground. No air resistance.
Measurements	30 noisy observations between 5° and 35°.
Original test	200 angles between 10° and 45°.

For the first pass, the analyst uses only the measurements.

The observed window

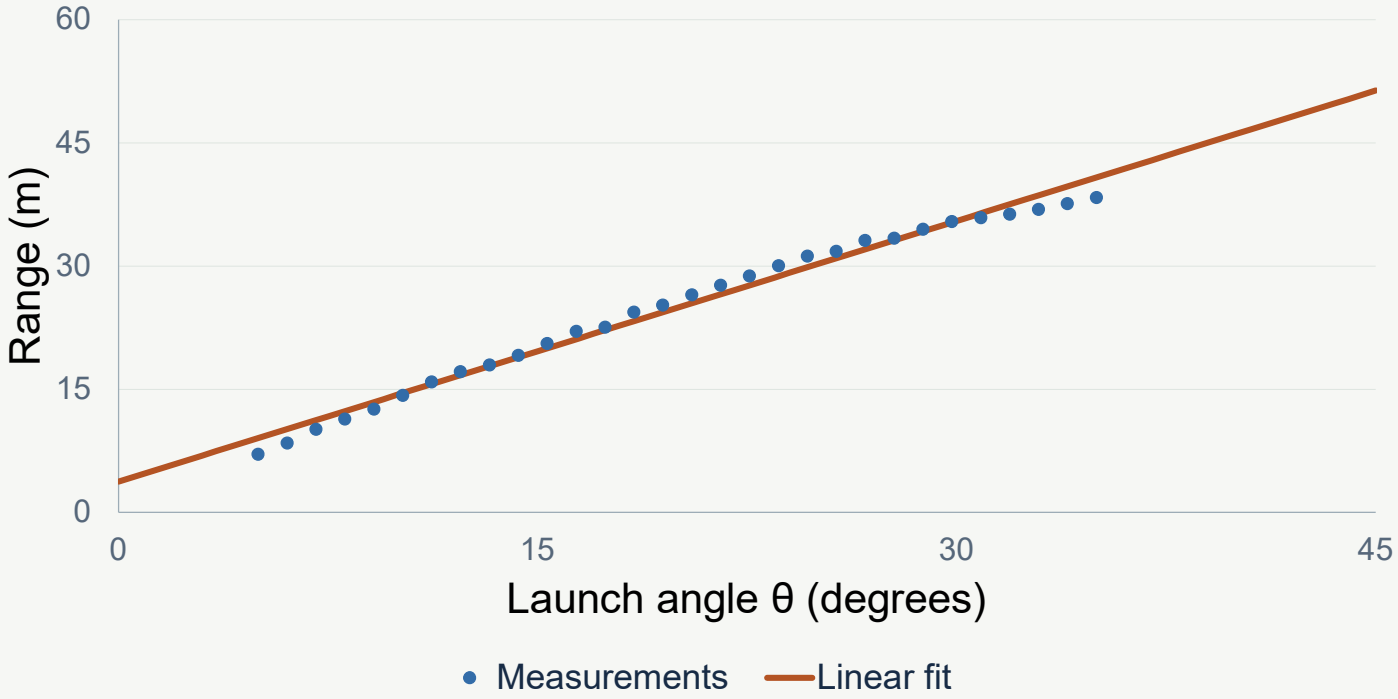


30 measurements

The observations cover 5°–35°.

Many functions could fit this window.

A straight-line baseline



$R^2 = 0.803$

Original test score 10°–45°.

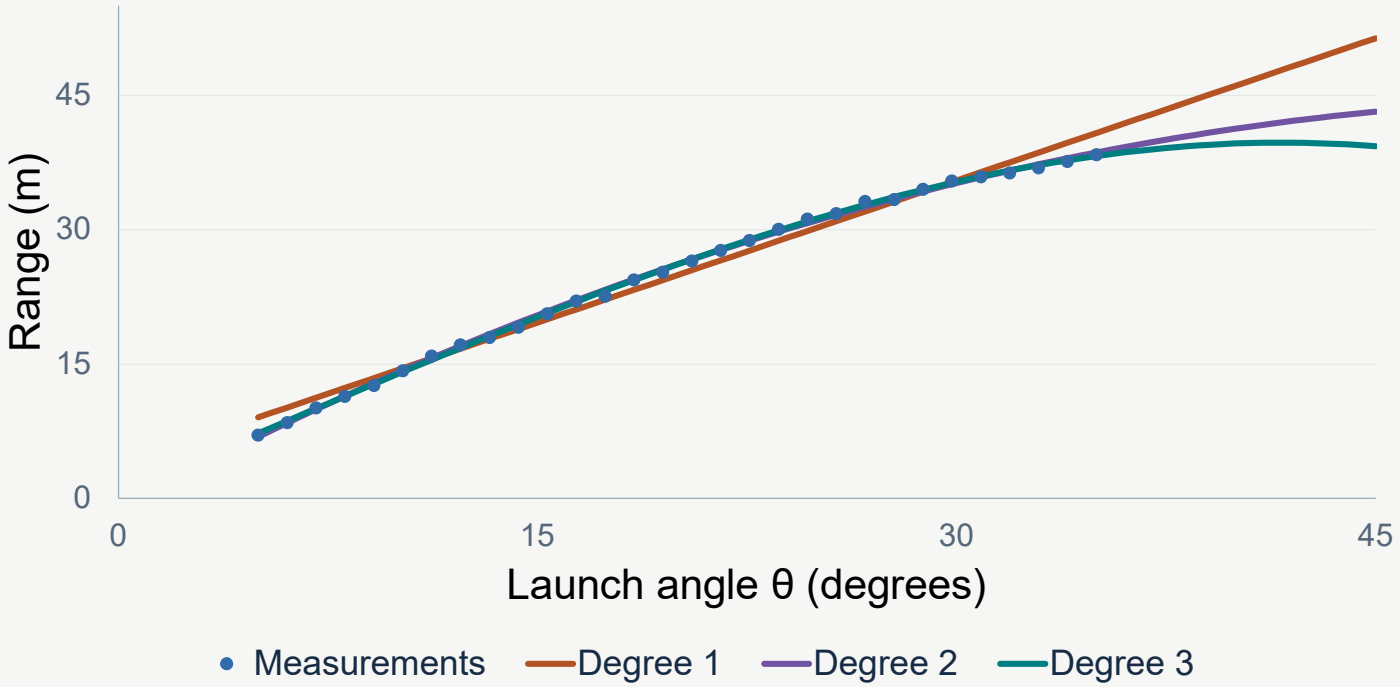
Training $R^2 = 0.986$.

Polynomial complexity changes the score

Degree	Training R^2	Test R^2
1	0.986	0.8027
2	0.999	0.9921
3	**0.999**	**0.9975**
5	0.993	−1.3687
12	0.796	−16763.7852

Higher degree can worsen both numerical stability and extrapolation.

Several curves fit the same window



Similar locally

Within 5°–35°,
the fits look close.

The curves separate
as the angle
increases.

Regularization changes the fit

Ridge α	Test R^2
0.001	0.9975
0.010	0.9975
1.000	0.9901
10.000	0.9133
100.000	0.7970

Degree-3 features

Changing the penalty alters coefficients.

It does not encode periodicity.

A model zoo on the original test

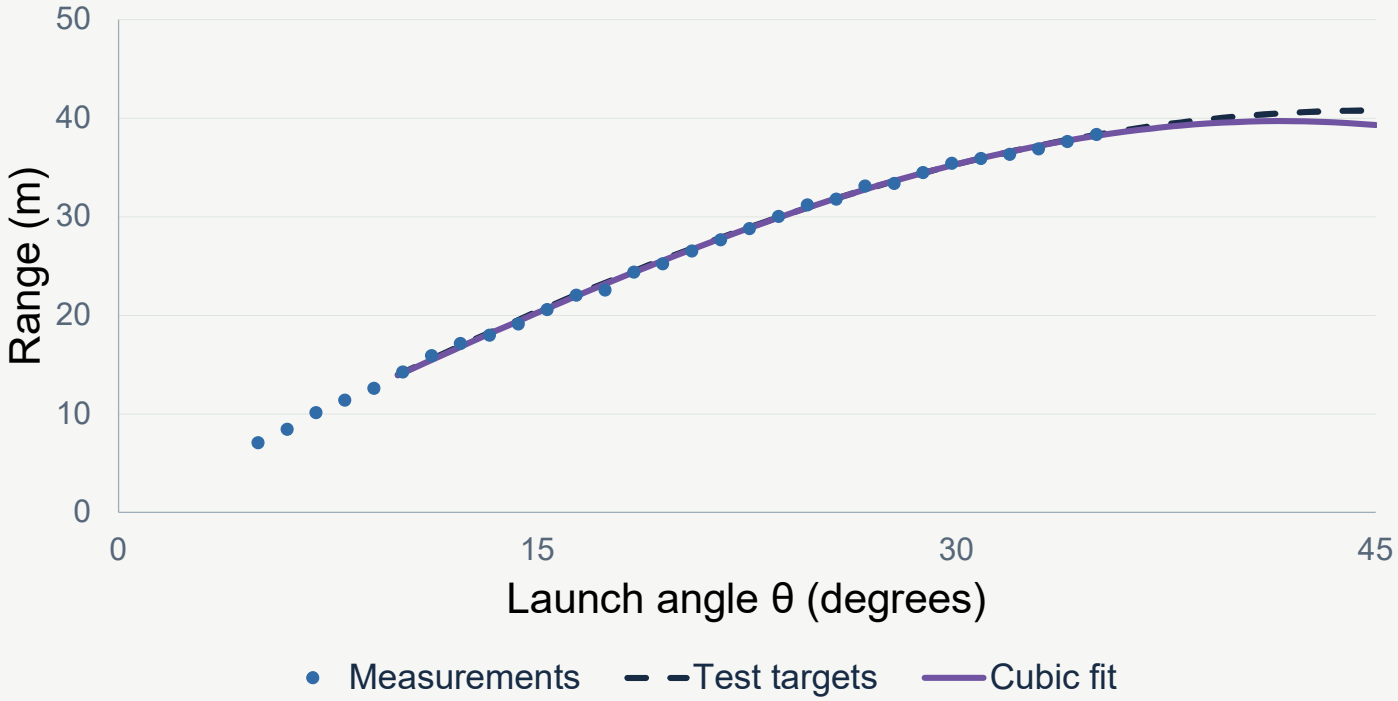
Model	Test R^2
Decision tree	0.985
Random Forest	0.981
Gradient boosting	0.9853
SVR	0.4591
k-NN	0.9598
MLP	0.3939

One test range

These scores compare performance over 10°–45°.

They do not describe behavior at 80°.

The apparent winner: a cubic model



$R^2 \approx 0.998$

The cubic model fits the original test very well.

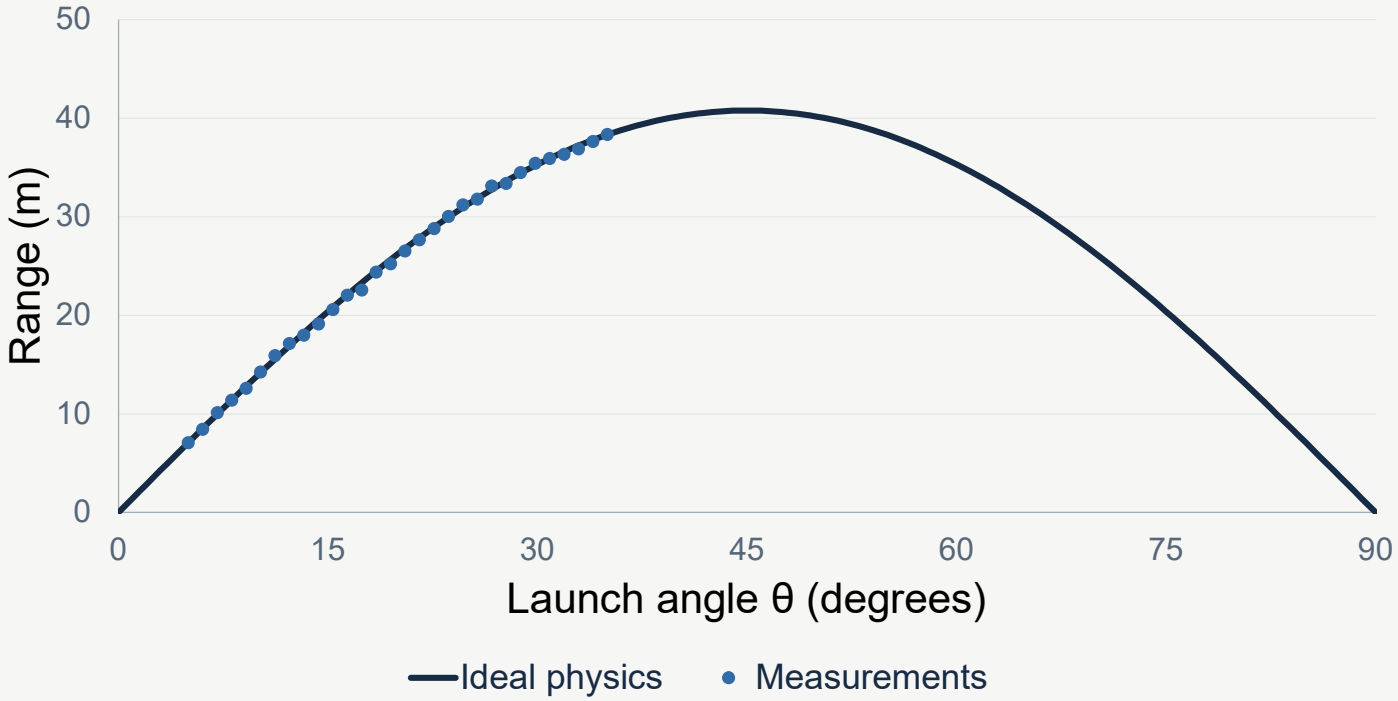
Original test: 10°–45°

PART 4

The debrief

What did the score leave untested?

The full physical shape

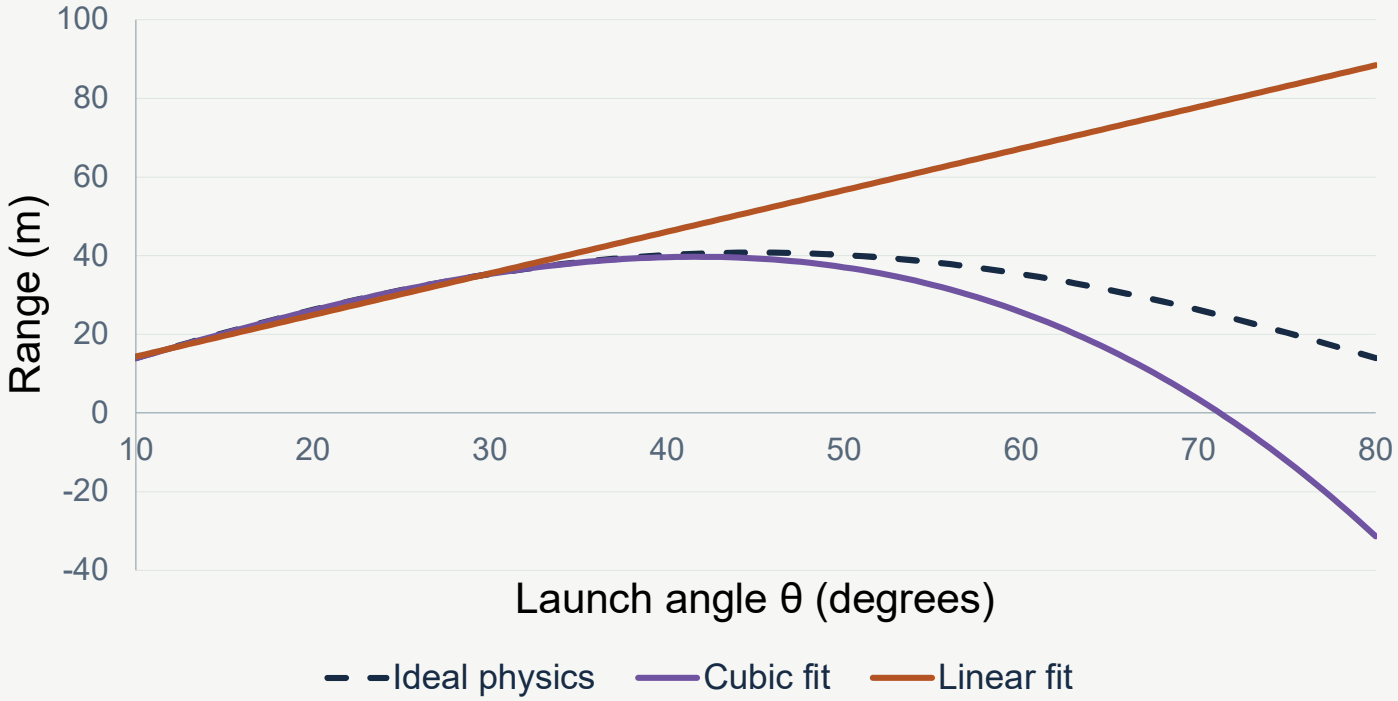


Maximum at 45°

$R(\theta) = (v^2/g) \sin 2\theta$

The range returns to zero at 90°.

Extrapolation exposes the mismatch



At 60°

True	35.31 m
Cubic	25.61 m
Linear	67.22 m

A wider test reveals different behavior.

Each model carries a shape assumption

Model family	Structure encoded
Linear regression	Straight line in the supplied features
Polynomial regression	An algebraic polynomial in the angle
Tree / Random Forest	Piecewise constant regions
k-NN	Local averaging around training examples
MLP	A nonlinear mapping set by its architecture

A locally accurate fit can still have the wrong global structure.

The cheap habit and its consequences

Silent failures	The chosen test can miss a shift or a spurious feature.
Weak justification	A high score does not justify the hypothesis class.
Blind improvement	Repeated tuning can replace diagnosis.
Poor transfer	A change in sensors or operating conditions can break the fit.

PART 5

A representation that respects the physics

The same measurements, expressed in a different feature space

Deriving the shape

$$x(t) = v\cos\theta \cdot t$$

$$y(t) = v\sin\theta \cdot t - \frac{1}{2}gt^2$$

$$t^* = 2v\frac{\sin\theta}{g}$$

Fixed speed. Constant gravity. Equal heights. No air resistance.

$$\boldsymbol{R(\theta)} = \left(\frac{\boldsymbol{v^2}}{\boldsymbol{g}}\right)\sin(\boldsymbol{2\theta})$$

A linear model in periodic features

$$\varphi(\theta) = (\sin 2\theta, \cos 2\theta)$$

$$R(\theta) = c + a \sin 2\theta + b \cos 2\theta$$

The features encode the period.
The fit estimates the scale, phase and intercept.

The feature transformation in code

```
def sin_cos_features(theta_deg):
    theta = np.deg2rad(theta_deg)
    return np.c_[np.sin(2 * theta), np.cos(2 * theta)]

Phi = sin_cos_features(X_train)
model = LinearRegression().fit(Phi, y_train)
```

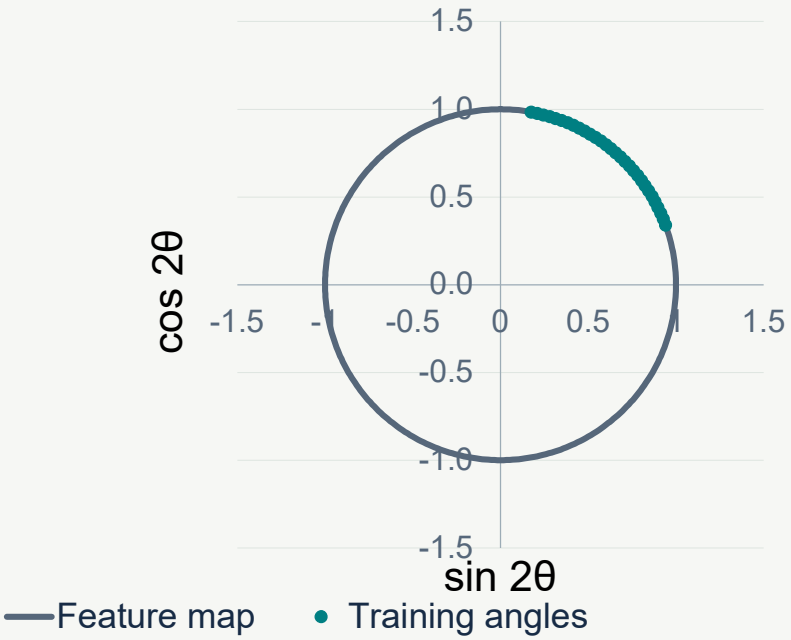
Same estimator

The representation changes.

The regression is still linear in its coefficients.

The default LinearRegression fit includes an intercept.

Periodic coordinates have a geometry

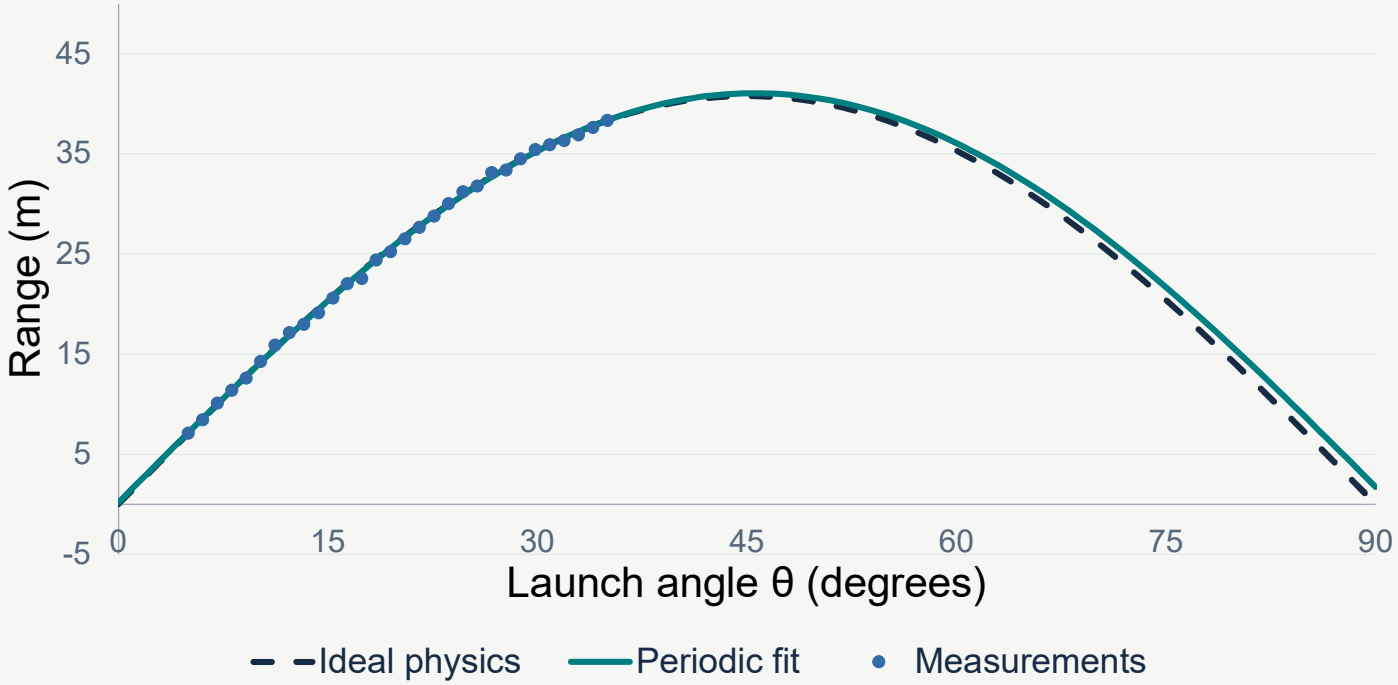


A unit circle

$\sin^2 2\theta + \cos^2 2\theta = 1$

The features encode periodicity before we fit the model.

The periodic model follows the shape



$R^2 = 0.9999$

Test: 10°–45°.

The wider plot shows extrapolation.

The same data at an unseen angle

60° lies beyond the 5°–35° training window.

At 60°	Range / prediction	Absolute error
Ideal physics	35.31 m	—
Linear fit	67.22 m	31.91 m
Cubic fit	25.61 m	9.71 m
Periodic fit	**36.05 m**	**0.74 m**

The useful change is the representation of the input.

The engineer’s responsibility

Formulate	Identify the phenomenon and the conditions where the model will operate.
Represent	Choose coordinates that preserve the structure needed for the task.
Diagnose	Test the model where its assumptions might fail.

Each algorithm earns its place by solving a representational problem.

PART 6

Exercises

We learn ML just through doing ML.

Exercises: shape and extrapolation

1 Hands-on	Write a complete version of our latest pipeline on your own. (PLEASE Don’t use LLMs)
2 Compare features	Try a single sine feature, then add a 3° phase shift. What changes?
3 Another shape	If $f(x) = \exp(-x^2)$, which features would express the expected structure?
4 A spurious feature	Build a high-scoring model that fails under a realistic distribution shift.

Session 1: the cantilever beam

How do we find the right representation when the useful coordinates are hidden?

A cantilever beam
A search for physical coordinates

SESSION 00

The representation is part of the model.

A defensible model needs assumptions that fit the phenomenon.